CARD 1 OF 8 DIESEL: HOW OFTEN **QUESTION**

How often does NFPA 25 want a **diesel** fire pump run on a **no-flow (churn)** test?

FSM · Fire System Mastery firesystemmastery.com

CARD 2 OF 8 DIESEL: HOW LONG **QUESTION**

During the weekly churn test, how long does a **diesel** pump have to run?

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CARD 3 OF 8 ELECTRIC: WEEKLY **QUESTION**

Most electric fire pumps can go monthly. Name the **four kinds that still need a weekly** churn test.

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CARD 4 OF 8 ELECTRIC: MONTHLY **QUESTION**

Which electric fire pumps are allowed to go to a **monthly** churn test?

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*FSM · Sprinkler Fact Cards — NFPA 25 (2020) Fire Pump Churn Tests · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 2 OF 8 DIESEL: HOW LONG **ANSWER**

At least 30 minutes.

Why so long: a diesel has to get up to operating temperature so you can watch the oil pressure, coolant and oil temperatures, and the heat exchanger actually doing their job.

A 5-minute bump proves it cranks. It doesn't prove it keeps running.

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On the job: start the clock when the engine is running, not when you hit the button. Log the start and stop times on the weekly report.

CARD 1 OF 8 DIESEL: HOW OFTEN **ANSWER**

Every week.

The one way out: an **approved risk analysis** can set a different frequency. Without one on file, it's weekly.

No-flow means the pump starts and runs with no water flowing out to a test header. It's proving the pump and driver start and run, not measuring flow.

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Trap: "We switched the diesel to monthly like the electric." Not without an approved risk analysis.

CARD 4 OF 8 ELECTRIC: MONTHLY **ANSWER**

Three routes:

- Any electric pump **not** on the weekly list.
- An electric pump system that has a **redundant fire pump**.
- Whatever frequency an **approved risk analysis** sets.

Monthly is permitted, not required. The owner can always test more often.

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On the job: before you drop a pump to monthly, check the four weekly cases first. A vertical turbine pulling from a pond is weekly no matter how the rest of the building is tested.

CARD 3 OF 8 ELECTRIC: WEEKLY **ANSWER**

1. Pumps serving a building **taller than the fire department can pump** to.
2. Pumps with a **limited service controller**.
3. Vertical turbine pumps.
4. Pumps drawing from a **ground-level tank** or any source with too little pressure to do much without the pump.

What they share: if the pump doesn't start, there's no real backup.

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Trap: "Electric pumps are monthly." Only the ones **not** on this list.

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 ELECTRIC: HOW LONG **QUESTION**

How long does an **electric** pump run on its churn test, and **how** do you start it?

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CARD 6 OF 8 WHO HAS TO BE THERE **QUESTION**

The controller has an automatic weekly timer. Does someone still have to be **at the pump** while it runs?

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CARD 7 OF 8 WHAT YOU RECORD **QUESTION**

The pump is running. Name **four things** you check or record during the churn test.

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CARD 8 OF 8 CHURN VS. ANNUAL **QUESTION**

The pump passes every weekly churn test. Is that enough for the **year**?

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CARD 6 OF 8 WHO HAS TO BE THERE **ANSWER**

Yes. Qualified personnel have to be in attendance whenever the pump runs. The timer doesn't change that.

The exception: remotely monitored automated testing done to the standard's rules. That includes automatic engine shutdown on a diesel, and someone qualified who can reach the pump within **5 minutes** if something goes wrong.

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Trap: "It's on a timer, nobody needs to go down there." A timer starts the pump. It doesn't watch it.

CARD 5 OF 8 ELECTRIC: HOW LONG **ANSWER**

At least 10 minutes.

Start it automatically, the way it would start in a fire: drop the pressure so the controller calls the pump. Pushing the start button skips the part you're trying to prove.

Automatic timer: allowed to replace the manual starting procedure, if it meets the standard's rules.

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Trap: 10 minutes is the **electric** number. Diesels are 30.

CARD 8 OF 8 CHURN VS. ANNUAL **ANSWER**

No. NFPA 25 also requires an **annual flow test**.

Churn test: proves the pump starts and runs. No water flows out.
Annual test: run at **no-flow (churn), rated flow and 150% of rated flow** through approved test devices. It proves the pump still delivers the water and pressure it was rated for.

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Trap: a pump can start every week for years and still fall short at rated flow. Only the annual test catches a worn impeller.

CARD 7 OF 8 WHAT YOU RECORD **ANSWER**

Any pump:
• The **starting pressure** (from the pressure switch or transducer).
• **Suction and discharge** gauge readings.
• Packing glands: a **slight** discharge is normal.
• Unusual **noise or vibration**.

Electric: time to reach full speed.
Diesel: time to crank, time to running speed, oil pressure, temperatures, and cooling water through the heat exchanger.

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On the job: a dry packing gland isn't a good sign. Packing is meant to drip a little to stay cool.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back